

PUID:

Due Sunday, October 12, 2025 at 11:59 pm

Instructions: There are 5 questions one of which is an extra credit. Please choose 4 questions to be graded and identify the bonus question. If no questions are selected, the first 4 will be graded. **The bonus question worth 5 points.** Time limit is 90 minutes.

1. [25 points] A class contains 10 boys and 9 girls. The teacher selects 5 of the children at random and without replacement. Find the probability that the number of girls selected exceeds the number of boys select.

Answer: Let $A = \{\# \text{ of girls} > \# \text{ of boys}\}$.

Cases are (0 boys and 5 girls) or (1 boy and 4 girls) or (2 boys and 3 girls).

$$\begin{aligned} P(A) &= \frac{|A|}{|\Omega|} = \frac{\binom{10}{0}\binom{9}{5} + \binom{10}{1}\binom{9}{4} + \binom{10}{2}\binom{9}{3}}{\binom{19}{5}} \\ &= \frac{9 \cdot 8 \cdot 7 \cdot 6 / 24 + 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 / 120 + 45 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot / 720}{19 \cdot 18 \cdot 17 \cdot 16 \cdot 15 / 120} \\ &= \frac{126 + 1,260 + 3,780}{11,628} = \frac{5,166}{11,628} = 0.4443. \end{aligned}$$

2. [25 points] A car safety institute observes that 15% of cars on the road are manufactured by brand “Purdue”. They also observe that 97% of “Purdue” brand vehicles are classified as safe at the time of inspection. The percentage of all cars (regardless of brand) classified as safe is 75%. A car is randomly chosen on the road and inspected. It is classified as safe. What is the conditional probability that it is a “Purdue” brand vehicle?

Answer: Let $A = \{\text{a car is “safe”}\}$. $B = \{\text{a car is manufactured by “Purdue”}\}$. We know that $P(A) = 0.75$, $P(A|B) = 0.97$, and $P(B) = 0.15$.

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A|B) \cdot P(B)}{P(A)} = \frac{0.97 \times 0.15}{0.75} = 0.194.$$

3. [25 points] An urn contains four fair dice:

- two have faces numbered 1,2,3,4,5,6;
- one has faces numbered 1,1,3,3,5,5;
- one has all six faces numbered 5.

One of the dice is randomly selected from the urn and rolled. The same die is rolled a second time. Calculate the probability that a 5 is rolled both times.

Answer: Let $A = \{5 \text{ is rolled both times}\}$, $B_1 = \{1, 2, 3, 4, 5, 6 \text{ dice are selected}\}$, $B_2 = \{1, 1, 3, 3, 5, 5 \text{ die is selected}\}$, and $B_3 = \{5, 5, 5, 5, 5, 5 \text{ dice are selected}\}$. Then, the total probability of A is

$$\begin{aligned} P(A) &= P(A \cap B_1) + P(A \cap B_2) + P(A \cap B_3) \\ &= P(A|B_1) \cdot P(B_1) + P(A|B_2) \cdot P(B_2) + P(A|B_3) \cdot P(B_3) \\ &= \left(\frac{1}{6}\right)^2 \cdot \left(\frac{2}{4}\right) + \left(\frac{2}{6}\right)^2 \cdot \left(\frac{1}{4}\right) + 1 \cdot \left(\frac{1}{4}\right) = 0.2917. \end{aligned}$$

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4. [25 points] Let X be a continuous random variable with PDF:

$$f_X(x) = \begin{cases} \frac{1}{30}x(1+3x), & 1 < x < 3; \\ 0, & \text{otherwise.} \end{cases}$$

Calculate $\text{Var}\left(\frac{1}{X}\right)$.

Answer:

$$E\left[\frac{1}{X}\right] = \int_1^3 \frac{1}{30} \cdot (1+3x) \, dx = \frac{1}{30} \left(x|_1^3 + \frac{3}{2}x^2|_1^3 \right) = \frac{1}{30} \left(2 + \frac{3}{2}(9-1) \right) = 0.4667$$

$$E\left[\frac{1}{X^2}\right] = \int_1^3 \frac{1}{30} \cdot \frac{1}{x}(1+3x) \, dx = \frac{1}{30} (\log x|_1^3 + 3x|_1^3) = \frac{1}{30} (\log 3 + 3 \cdot 2) = 0.2366$$

Therefore,

$$\text{Var}\left[\frac{1}{X}\right] = 0.2366 - (0.4667)^2 = 0.0188.$$

5. [25 points] Let X be a continuous random variable with PDF:

$$f_X(x) = \begin{cases} 2x^{-3}, & x > 1; \\ 0, & \text{otherwise.} \end{cases}$$

Calculate $\mathbb{P}\left(\left|X - \frac{1}{2}\right| > 0.6\right)$.

[Answer:](#)

$$\begin{aligned} \mathbb{P}\left(\left|X - \frac{1}{2}\right| > 0.6\right) &= \mathbb{P}\left(\left\{X - \frac{1}{2}\right\} < -0.6 \cup \left\{X - \frac{1}{2}\right\} > 0.6\right) \\ &= \mathbb{P}(X < -0.1) + \mathbb{P}(X > 1.1) = 0 + \mathbb{P}(X > 1.1) \\ &= \int_{1.1}^{\infty} 2x^{-3} dx = 2 \frac{1}{-2} x^{-2} \Big|_{1.1}^{\infty} = (-1) \cdot (0 - (1.1)^{-2}) = \left(\frac{10}{11}\right)^2 = 0.8265. \end{aligned}$$

Good Luck!